

3/12/25

PH3102 Quantum Mechanics
Instructor: Dr. Siddhartha Lal
Expected End-sem Paper
Autumn Semester, 2025

Date: 3rd December, 2025
Full marks: 50
Time: 3 hours

Instructions: Answer all questions for full marks. Put your name and roll no. on both Question paper as well as the Answer script, and hand in both of them.

1 Variational Method for Quartic Potential.

[12 marks]

Consider a particle of mass m moving in a one-dimensional potential well given by $V(x) = \lambda x^4$, where $\lambda > 0$ is a constant with dimensions of energy/length⁴.

- (a) Write down the Hamiltonian H for the system. Using a Gaussian trial wavefunction of the form:

$$\psi_\alpha(x) = A e^{-\alpha x^2}$$

determine the normalization constant A in terms of the variational parameter α . [3 marks]

- (b) Calculate the expectation values of the kinetic energy $\langle T \rangle$ and the potential energy $\langle V \rangle$ with respect to the trial wavefunction $\psi_\alpha(x)$.

(Hint: You may use the standard integral $\int_{-\infty}^{\infty} x^{2n} e^{-\beta x^2} dx = \frac{(2n-1)!!}{2^n \beta^n} \sqrt{\frac{\pi}{\beta}}$) [5 marks]

- (c) Minimise the total variational energy $E(\alpha) = \langle T \rangle + \langle V \rangle$ with respect to the parameter α to find the optimal α_0 . Using this, calculate the upper bound on the ground state energy E_0 . [4 marks]

2 Perturbation Theory on a Spin System.

[10 marks]

Consider a spin-1/2 particle characterized by the unperturbed Hamiltonian $H_0 = \omega S_z$, where ω is a constant frequency. Suppose the system is subjected to a small perturbation given by $H' = g S_x$, where $g \ll \omega$.

- (a) Write down the exact energy eigenvalues $E_n^{(0)}$ and eigenstates $|\psi_n^{(0)}\rangle$ (in terms of the standard basis $|\uparrow\rangle, |\downarrow\rangle$) of the unperturbed Hamiltonian H_0 . [2 marks]

- (b) Using second-order time-independent perturbation theory, calculate the energy correction $E_0^{(2)}$ to the ground state. [4 marks]

- (c) Calculate the first-order correction to the ground state wavefunction $|\psi_0^{(1)}\rangle$. Express your answer as a linear combination of the unperturbed eigenstates. [4 marks]

3 Interacting Spins and Magnetic Field.

[12 marks]

Two spin-1/2 particles, S_1 and S_2 , interact with each other via a Heisenberg exchange coupling J and are placed in a uniform external magnetic field B along the z-axis. The Hamiltonian is given by:

$$H = J(\vec{S}_1 \cdot \vec{S}_2) + B(S_{1z} + S_{2z})$$

Assume $J > 0$ (antiferromagnetic coupling).

- (a) Express the dot product operator $\vec{S}_1 \cdot \vec{S}_2$ in terms of the total spin operator S^2 and the individual spin operators S_1^2 and S_2^2 . [3 marks]

- (b) Find the exact energy eigenvalues for the four basis states of the coupled system: the Triplet states ($|1, 1\rangle, |1, 0\rangle, |1, -1\rangle$) and the Singlet state ($|0, 0\rangle$). [5 marks]
- (c) Sketch the energy levels as a function of the magnetic field B . Identify if any level crossing occurs (i.e., does the ground state change nature as B increases?). Determine the critical field B_c where this transition occurs. [4 marks]

4 Density Matrix dynamics.

[10 marks]

A quantum system is described by a mixed state density matrix:

$$\rho = \frac{3}{4}|\uparrow\rangle\langle\uparrow| + \frac{1}{4}|\downarrow\rangle\langle\downarrow|$$

where $|\uparrow\rangle$ and $|\downarrow\rangle$ are eigenstates of σ_z .

- (a) Calculate the purity of this state, defined as $\text{Tr}(\rho^2)$. Based on this value, confirm whether the state is pure or mixed. [2 marks]
- (b) Calculate the expectation value of the spin operator $\langle S_x \rangle$ for this system. [3 marks]
- (c) If the system evolves under the Hamiltonian $H = \omega S_z$, find the density matrix $\rho(t)$ at a later time t using the Liouville-von Neumann equation (or the time evolution operator). Does the purity of the state change with time? [5 marks]

5 Landau Levels.

[6 marks]

Consider a charged particle of charge q and mass m moving in the xy -plane subject to a uniform magnetic field $\vec{B} = B\hat{k}$. The Hamiltonian is given by $H = \frac{1}{2m}(\vec{p} - q\vec{A})^2$.

- (a) Using the symmetric gauge $\vec{A} = \frac{1}{2}(-By, Bx, 0)$, construct the appropriate ladder operators a and $a^\dagger$ that diagonalize this Hamiltonian. State the commutation relation $[a, a^\dagger]$. [3 marks]
- (b) Write down the expression for the energy spectrum (Landau Levels) E_n in terms of the cyclotron frequency ω_c . State the degeneracy of the ground state. [3 marks]